3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Dual Source Time-of-Flight Mass Spectrometer and Sample Handling System

Authors & Institutes:
• William B. Brinckerhoff, Timothy J. Cornish — *Johns Hopkins University/Applied Physics Laboratory*
• P. R. Mahaffy — *NASA/Goddard Space Flight Center*

ABSTRACT

We present details of an instrument under development for potential NASA missions to planets and small bodies. The instrument comprises a dual ionization source (laser and electron impact) time-of-flight mass spectrometer (TOF-MS) and a carousel sample handling system for in situ analysis of solid materials acquired by, e.g., a coring drill. This "DS-TOF" instrument could be deployed on a fixed lander or a rover, and has an open design that would accommodate measurements by additional instruments. The sample handling system (SHS) is based on a multi-well carousel, originally designed for a comet mission. Solid samples, in the form of drill cores or as loose rock chips or fines, are inserted into the SHS through an access port, sealed in vacuum, and transported around the carousel to a pyrolysis cell and/or directly to the TOF-MS inlet. Samples at the TOF-MS inlet are xy-addressable for laser or optical microprobe. Cups may be ejected from their holders for analyzing multiple samples or caching them for return. Samples are analyzed with laser desorption and evolved-gas/electron-impact sources. The dual ion source permits broad studies of elemental, isotopic, and molecular composition of unprepared samples with a single mass spectrometer. Pulsed laser desorption permits the measurement of abundance and isotope ratios of refractory elements, as well as the detection of high-mass organic molecules in solid samples. Evolved gas analysis permits similar measurements of the more volatile species in solids and aerosols. The TOF-MS is based on previous miniature prototypes at JHU/APL that feature high sensitivity and a wide mass range. The laser mode, in which the sample cup is directly below the TOF-MS inlet, permits both ablation and desorption measurements, to cover elemental and molecular species, respectively. In the evolved gas mode, sample cups are raised into a small pyrolysis cell and heated, producing a neutral gas that is electron ionized and pulsed into the TOF-MS. (Any imaging and laser microprobe studies would necessarily precede the pyrolysis step to assure that the grain-scale composition is captured.) Details will be given on the DS-TOF operating modes envisioned for various in situ measurement objectives on asteroid and Mars missions. Initial results from the laboratory breadboard effort, including recent analyses of various calibration samples, SHS sealing mechanisms, and vacuum system requirements (for Mars) will be presented. The authors acknowledge the contributions of A. Cheng, H. Niemann, D. Harpold, D. Yucht, S. Rafeek, S. Gorevan, S. Ecelberger, T. McCoy, E. Vicenzi, and G. Managadze. Support is provided by NASA Grant NAG5-10693.